A PROJECT PRESENTATION ON

SPEED AND DIRECTION CONTROL OF STEPPER MOTOR USING IGBT

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ABSTRACT

- A stepper motor rotates in precise steps (1.8 degrees per step), used in applications requiring accurate movement, like robotics and antennas. It can be unipolar or bipolar.
- The project uses **ULN2003 IC** and **L293D Motor Driver** to drive the stepper motor, as they provide the necessary current that the controller alone cannot supply.
- The motor is controlled via **buttons**: a **Forward Button** for clockwise rotation, a **Backward Button** for counterclockwise rotation, and a **Stop Button** to halt the movement.
- Half-Stepping is used to achieve smoother and more precise motor rotation by incrementally stepping through positions.

REVIEW OF LITERATURE

Stepper motors are essential in precision control applications like robotics, CNC machines, and 3D printers due to their ability to divide full rotations into discrete steps, enabling high-precision control of position and speed without feedback systems. This is achieved by energizing the motor's coils in a specific sequence to control rotation, speed, and direction effectively.

Key Points:

- **Stepper motors** offer precise, step-based motion control without feedback systems.
- **Raspberry Pi** controls motors via GPIO pins using methods like full-step, half-step, and micro stepping.
- **Speed and direction** are managed by pulse timing and signal sequencing (or PWM).
- **IGBTs** act as efficient, high-power switches in motor control circuits.
- **Pi + IGBT integration** enables low-voltage control of high-power motors with better efficiency.

INTRODUCTION

- **Stepper motors** are ideal for precise motion control in applications like robotics and 3D printing due to their stepwise rotation and ability to work without feedback systems.
- The **Raspberry Pi** offers a flexible and low-cost platform to control stepper motors using GPIO pins and libraries, supporting techniques like micro stepping and full/half-step driving.
- **Speed and direction** are controlled by adjusting pulse frequency and signal sequence. PWM from Raspberry Pi helps manage motor speed more smoothly.
- **IGBTs (Insulated Gate Bipolar Transistors)** are efficient power switches that handle high voltage and current, making them suitable for controlling large or fast stepper motors.
- **Combining Raspberry Pi with IGBTs** enables precise, low-voltage control of high-power stepper motors, improving efficiency, reducing heat, and enhancing overall system performance

PROBLEM FORMATION

- IGBTs need careful handling because they work with high voltage and current.
- No feedback system, so the motor may lose steps or go off position.
- Heat can build up in IGBTs without proper cooling.
- Raspberry Pi may not give perfectly timed signals for the motor.
- Circuit may lack protection parts like diodes to handle motor back current.
- There' s no current control, which can damage the motor or IGBT.
- PWM usage isn' t fully explained, which is important for smooth speed contro

EXISTING SYSTEM

stepper motor control is generally achieved using **basic switching devices** such as **BJTs (Bipolar Junction Transistors)** or **MOSFETs (Metal-Oxide Semiconductor Field-Effect Transistors)**. These systems are typically controlled using **microcontrollers or microprocessors**, generating pulse signals to drive the motor.

- ► **Simple Switching Devices**
- ► **Open-Loop Control**
- ► **Limited Speed Control**
- ► **Direction Control**
- ► **Lower Efficiency**

❖These limitations highlight the need for a simpler, more efficient, and cost effective solution like PWM-based control.

PROPOSED SYSTEM

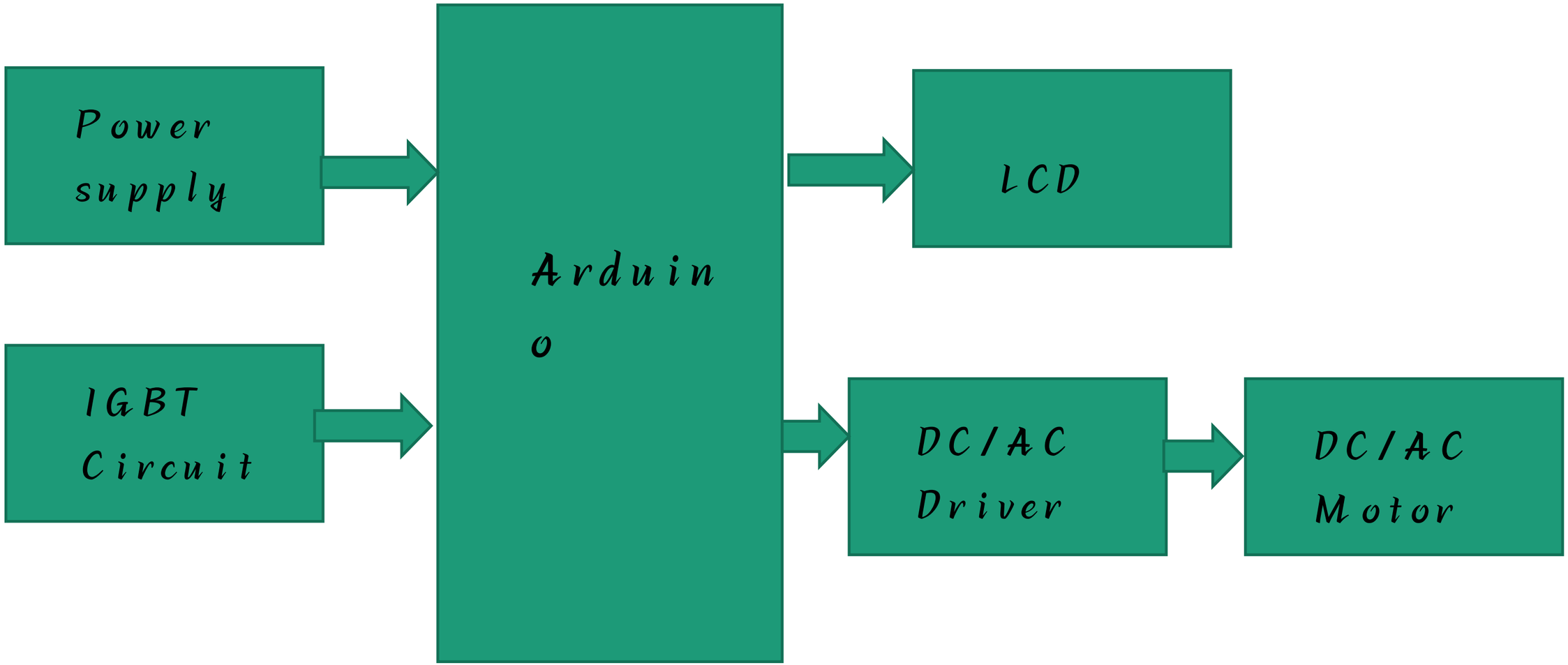
To design a system that controls the **speed** and **direction** of a **stepper motor** precisely using **Insulated Gate Bipolar Transistors (IGBTs)** for efficient switching and power handling.

- ► **Microcontroller / Controller (e.g., Arduino / Raspberry Pi / 8051 / STM32)**
- ► **IGBT Driver Circuit**
- ► **IGBTs**
- ► **Stepper Motor (Unipolar or Bipolar)**
- ► **Power Supply**
- ► **User Interface**

OBJECTIVES

- IGBTs offer high voltage capability, low ON-resistance, fast switching, easy drive, and require no gate current.
- Stepper motors are increasingly used due to being economical and easy to control.
- Low-cost microcontrollers have made stepper motor control more flexible and versatile.
- Stepper motors are easy to interface with digital components.
- They allow open-loop position control by moving one step per input pulse.
- No analog-to-digital conversion is needed, unlike with AC or DC motors.
- The project aims to build a microcontroller-based controller for smooth speed and position control of stepper motors

DESIGN METHODOLOGY



HARDWARE KIT DESGIN



EXPERIMENTAL RESULT

- Stepper motor can be termed as digital motor because it operates on pulses. Unlike AC or DC motor that rotates continuously, stepper motor rotates in steps. It rotates in number of steps as per applied number of pulses.
- We demonstrate and integrated system which consists of electrical, electronics, programming (encryption and coding). Raspberry pi was used to operate and control the switching actions of dc motor and also stepper motor. The speed of dc motor and stepper motor varies

APPLICATIONS

- **CNC Machines:**
  - Stepper motors are used for precise tool movement and position control in milling, cutting, and drilling machines.
- **3D Printers:**
  - Enables accurate control of printing head movement and build platform adjustment.
- **Robotics:**
  - Used in robotic joints and wheels for accurate, repeatable motion control.
- **Conveyor Belt Systems:**
  - Controls the speed and direction of conveyor belts in industries for material handling

CONCLUSION

- A stepper motor can rotate by directly connecting it to a power supply.
- Using a Python program with Raspberry Pi allows control over the motor's rotation.
- Speed and direction can be adjusted by modifying the Python code.
- This method is especially useful for industrial automation processes.
- Stepper motors have many applications, but programming enhances their use in automation

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THANK YOU

SPEED AND DIRECTION CONTROL OF STEPPER MOTOR USING IGBT

We appreciate your time and attention during this presentation. Your feedback and support are highly valued

Presented by: Team 05

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